Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 10
Chapter 13: Statistics

Time: 60 Minutes
MM: 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. The median of a frequency distribution is found graphically using: [1]
(A) Histogram (B) Frequency polygon (C) Ogives (D) Pie chart

Q2. If the mean of 6 observations is 12 and each observation is divided by 2, then the new mean will be: [1]
(A) 12 (B) 6 (C) 24 (D) 3

Q3. In a continuous frequency distribution, the modal class is $40 - 60$. If $l = 40, f_1 = 20, f_0 = 12, f_2 = 11, h = 20$, the mode is: [1]
(A) 48.42 (B) 49.41 (C) 50.25 (D) 47.85

Q4. The cumulative frequency of the last class interval in a cumulative frequency table is equal to: [1]
(A) the class mark of the last class (B) the frequency of the first class
(C) the total frequency of the distribution (D) the range of the distribution

Q5. If the mean of first n odd natural numbers is 15, then n is equal to: [1]
(A) 15 (B) 30 (C) 225 (D) 14

Q6. **Assertion (A):** If the mean of five numbers is 20, and one number is excluded, their mean becomes 18, then the excluded number is 28.
Reason (R): Excluded value is computed by subtracting the new total sum from the original total sum. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Find the arithmetic mean of the following discrete frequency distribution: [2]

x	10	20	30	40
f	3	2	4	1

Q8. If the mode of a frequency distribution is 24 and the mean is 27, find its median using empirical relation. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Convert the following 'more than' distribution into regular class intervals and calculate the mean: [3]

Production	≥ 0	≥ 10	≥ 20	≥ 30	≥ 40
Farms	50	46	38	20	8

Q10. Find the modal lifetime of 225 electrical components: [3]

Lifetime (hours)	0–20	20–40	40–60	60–80	80–100	100–120
Components	10	35	52	61	38	29

SECTION D: Long Answer Type (2 × 5 = 10 Marks)

Q11. The lengths of 40 leaves of a plant are measured correct to the nearest millimetre. Convert the data into continuous classes and calculate the median length: [5]

Length (mm)	118–126	127–135	136–144	145–153	154–162	163–171	172–180
Leaves	3	5	9	12	5	4	2

Q12. Find the missing frequencies x and y in the following distribution table, given that the median is 32 and the total frequency is 100: [5]

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	10	x	25	30	y	10

SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

Q13. CASE STUDY: COMMERCIAL LOGISTICS FLEET FUEL EFFICIENCY BENCHMARKING [4 Marks]

A freight logistics enterprise tracked the operational fuel mileage (km per litre) of 60 heavy-duty cargo trucks operating across national supply corridors.

Mileage (km/l)	10–12	12–14	14–16	16–18	18–20
Truck Count	7	12	18	13	10

(i) Calculate the arithmetic mean fuel efficiency of the commercial fleet. [2]

(ii) State the modal fuel efficiency class and calculate its mode. [1]

(iii) What percentage of trucks achieved an optimal mileage of 16 km/l or higher? [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★